

LAUREN LIAO

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EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS

Pittsburgh, PA/USA

Master of Science in Computational Finance – MSCF

Expected 12/26

- Best Risk Management Award in RITC x CMU 2025 Trading Competition - volatility & algo cases

UNIVERSITY OF WATERLOO, MATH & BBA DOUBLE DEGREE PROGRAM

Waterloo, ON/Canada

Bachelor of Mathematics – Statistics and Mathematical Optimization Double Major **GPA: 4.0/4.0**

4/22

- Participated in Poker club and Investment club activities

Bachelor of BBA at Wilfrid Laurier University – Finance Specialization **GPA: 3.9/4.0**

EXPERIENCE

TD SECURITIES

Toronto, ON

Front Office Quant, Fixed Income Derivatives Strategist (5/24 – 7/25)

5/21 – 8/21; 9/22 – 7/25

- Implemented valuation engines for treasury locks and total return swaps in C++; updated library based on business needs
- Developed and supported x-gamma risk tool for trading desks, involving python/vba programming and UI development
- Created Flask and Streamlit app using asynchronous programming in python to centralize and automate daily tasks
- Proposed and built CI/CD pipelines using GitHub Actions/Jenkins to streamline deployment of pricing models and unit tests

Global Markets Rotating Associate (9/22 – 4/24)

- *Equities Algo MM* - Researched on Hedge Fund activities and developed reactive strategies; Utilized Random Forests model to predict position imbalance; Researched on EOD market impacts for CRB using historical data, conducted factor analysis
- *Structured Notes Trading* - Created Dash app to visually analyze shocks & spreads, resulting in 60% in pricing efficiency
- *IRS Trading* - Gained strong understanding of swaps trading, hedging strategies, and risk-taking side of business
- *XVA* - Improved capital management efficiency by creating top of house view of counterparty risks and RAROC sheet

Summer Associate –ETF Trading (5/21 – 8/21)

- Backtested and analyzed ETF automated trading strategies utilizing Onetick, Excel, SQL and Python
- Generated weekly ETF Strategy reports of market focus and fund flows, streamlined processes to python, saving 50% time

PURPOSE INVESTMENTS

Toronto, ON

Quantitative Analyst Intern

1/21 – 4/21

- Developed visual market analytics on company website using Pandas, Plotly, REST API and Git
- Researched on covered call strategies and reverse mortgage by backtesting in Python and conducting quantitative analysis

RBC CAPITAL MARKETS

Toronto, ON

Financial Analyst Intern – Product Control and Valuations

1/20 – 4/20

- Performed analysis for mark to market of the portfolio and residuals for UK Credit and Rates derivatives trading books

COMPETITIONS

Global Top 12 Team - Rotman International Trading Competition (RITC)

2/21

- Developed automated trading strategies using RIT API and Python to react to rapidly changing markets
- Executed delta-hedged and volatility arbitrage strategies, balancing risk exposures while maximizing P&L

Top 15 Finalist - National Undergraduate Big Data Challenge

5/20 - 7/20

- Applied Machine Learning algorithms (tensorflow, sklearn) and forecasting models (non-seasonal ARIMA, cross correlation) in Python and R to identify key factors for COVID-19 outbreak and predict future trends

ADDITIONAL INFORMATION

- Certificates: **Chartered Financial Analyst (CFA)**
- Programming skills: Python/VBA/SQL (proficient - numpy, pandas, sklearn, asynchronous programming), C++ (intermediate)
- Interests: Running half-marathons, Squash, Reading, Playing pipa, swimming